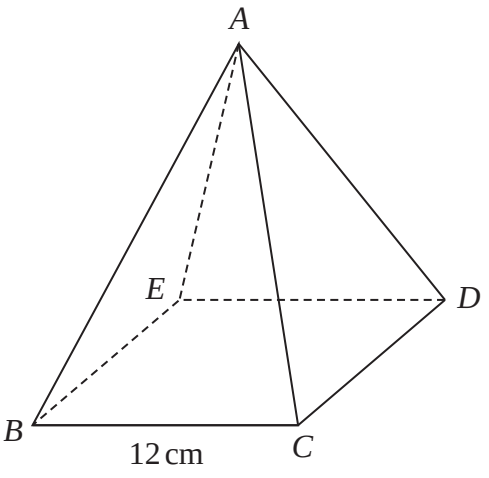
21 The diagram shows the solid pyramid *ABCDE*.



The base *BCDE* of the pyramid is a horizontal square with sides of length 12 cm.
The vertex *A* of the pyramid is vertically above the midpoint of *BD*.

The volume of the pyramid is 768 cm^3

Calculate the size of the angle between *AB* and the base of the pyramid.
Give your answer correct to 1 decimal place.

Let M be the midpoint of diagonal BD.

$768 = (1/3) \times 12^2 \times AM$

1 mark
Pyramid volume: one third × base area × vertical height.

$AM = 768 \times 3 \div 144 = 16$ cm

$BD = \sqrt{(12^2 + 12^2)} = 12\sqrt{2}$ cm

1 mark
Half the square diagonal is BM.

$BM = BD \div 2 = 6\sqrt{2}$ cm

tan angle ABM = AM ÷ BM = 16 ÷ (6√2)

Angle between AB and the base = 62.1°

2 marks total
Use the right triangle ABM: opposite16, adjacent6√2.

62.1 °

(Total for Question 21 is 4 marks)

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